MTH 200 **Derivatives & Notation**

Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer each and every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

BEFORE YOU START

Remember that the notation $\frac{dy}{dx}$ is an alternative way of expressing the derivative of a function y.

This notation is actually made up of two parts: $\frac{d}{dx}$ & y. Here, $\frac{d}{dx}$ acts as an "operator" meaning that it is notation that is meant to indicate that some computation is to be done and y acts as the thing this operation is being performed on. You can think of this much like how you would the notation. We know that $\sqrt{}$ is just a symbol meant to indicate an operation but is meaningless by itself. The notation $\sqrt{y}$ means that we will be performing the operation $\sqrt{}$ to y. The notation $\frac{dy}{dx}$ or $\frac{d}{dx}[y]$ has the same meaning.

$$\frac{dy}{dx} = \begin{array}{l} \leftarrow \text{The letter on top is the name of function being differentiated.} \\ \leftarrow \text{The letter on bottom is the variable being differentiated with "respect to"} \\ \qquad\qquad \text{(all other variables are considered constants)} \end{array}$$

Challenge Questions

1. Consider the functions $y = x^2$, $f(x) = \frac{1}{4}x^4 + 5x$, $g(t) = (2t+3)^3 + \pi$.

a) Compute $\frac{dy}{dx}$.

b) Compute $\frac{d}{dx}[y]$.

c) Compute $\frac{dy}{dx}y$.

d) Compute $\frac{df}{dx}$.

e) Compute $\frac{d}{dx}\left[f'(x)\right] \cdot f(x)$.

f) Compute $\frac{dg}{dt}$.

g) Compute $\frac{dg}{dt} g(t)$.

h) Compute $\frac{dg}{dx}$.

i) Compute $\frac{dy}{dt}$.

It is important to remember that our derivative "shortcuts" come from the definition of the derivative. This definition is important anytime we want to determine a derivative for which we don't have a shortcut. It also is important to remember that for a function to be differentiable at a point $x = a$ this means we need $f'(a)$ to exist and this implies that the limit in the definition of the derivative must exist.

2. Determine the derivative of $f(x) = 2x - 3$ by using the definition of the derivative *and* derivative shortcut rules.

3. When finding the derivative of $f(x) = 2x - 3$ using shortcut rules you needed to use the *Constant Rule*, the *Constant Multiple Rule*, and the *Difference Rule*. Use the definition to prove these rules.

It is also important not to lose sight of what a *derivative* represents and where they come from. With this in mind, consider the following questions.

4. The function f has the property that $f(4)=1$ and the tangent line to its graph at $x=4$ passes through the point $(-2,-3)$.

a) What is $f'(4)$?

b) If $h(x)=9f(x)-x^2$, find $h'(4)$.

5. Consider the function $f(x)=|x|$ and recall that this function is actually defined as a piecewise function

$$f(x)=|x|=\begin{cases} x & if \ \ x\geq 0 \\ -x & if \ \ x<0 \end{cases}.$$

a) Is $f(x)$ continuous at $x=0$? Justify your answer.

b) Use the definition of the derivative to determine if $f(x)$ is differentiable at $x = 0$.

c) What does this example tell you about the relationship between continuity and differentiability?